

2. (1 pt) The change in position is:

Slope between two points on the position-time graph	Area under a position-time graph	Slope between two points on the velocity-time graph	Area under the velocity-time graph
☐	☐	☐	☒

3. (1 pt) A particle starts from the position
- $(8\hat{i} + 5\hat{j})\text{ft}$
- with an initial velocity of
- $(3\hat{i} - 4\hat{j})\text{ft/s}$
- and an acceleration of
- $(-\hat{i} + 2\hat{j})\text{ft/s}^2$
- . The object reaches the maximum x-position when

$v_{\text{total}} = 0$	$v_x = 0$	$v_y = 0$
☐	☒	☐

4. (1 pt) A basketball is falling at terminal speed. What can be said about the drag force?

Drag Force < Weight	Drag Force = Weight	Drag Force > Weight
☐	☒	☐

5. (1 pt) A tennis ball is thrown through the air. The direction of the air resistance force is:

in the direction of the position vector	opposite the direction of the position vector	in the direction of the velocity vector	opposite the direction of the velocity vector	in the direction of the acceleration vector	opposite the direction of the acceleration vector
☐	☐	☐	☒	☐	☐

6. (1 pt) The impulse is determined as:

Area under force-displacement graph	Area under force-time graph	Area under force-acceleration graph
☐	☒	☐

7. (1 pt) A basketball is thrown across the gym so that it bounces on the floor a couple of times. Assuming no air resistance and no friction, the velocity of the basketball parallel to the floor:

changes only while the ball is in the air	changes only while the ball is bouncing	changes both when the ball is in the air and when it is bouncing	does not change
☐	☐	☐	☒

8. (1 pt) Angular momentum can be calculated as:

mv	mvr	ma	mar	mv^2	$0.5mv^2$
☐	☒	☐	☐	☐	☐

9. (1 pt) An object is spinning at 7 rad/sec. The angular speed of the object in rpm is:

0.0186 rpm	0.183 rpm	0.733 rpm	66.8 rpm	660 rpm	2640 rpm
☐	☐	☐	☒	☐	☐

10. (3 pts) Convert the acceleration of 3 ft/s^2 to units of block/NYsecond/fortnight.

1 block = 264 ft

1 NYsecond = 1 millisecond (the period of time in New York City between the traffic light turning green and the cab behind you honking)

1 fortnight = 1,209,600 seconds

$$13.7 \frac{\text{block}}{\text{NYsec} \cdot \text{fortnight}}$$

$$3 \frac{\text{ft}}{\text{s}^2} \cdot \frac{1 \text{ block}}{264 \text{ ft}} \cdot \frac{0.001 \text{ sec}}{1 \text{ NYsec}} \cdot \frac{1209600 \text{ sec}}{1 \text{ fortnight}} = 13.74$$

11. (3 pts) Determine the magnitude of the vector $\vec{A} - 2\vec{B}$, where $\vec{A} = (23\hat{i} + 64\hat{j})\text{ft}$ and $\vec{B} = (-16\hat{i} + 12\hat{j})\text{ft}$.

$$68.0 \text{ ft}$$

$$\begin{aligned} \vec{A} - 2\vec{B} &= (23 - 2(-16))\hat{i} + (64 - 2(12))\hat{j} \\ &= 55\hat{i} + 40\hat{j} \end{aligned}$$

$$|\vec{A} - 2\vec{B}| = \sqrt{(55)^2 + (40)^2} = 68.01 \text{ ft}$$

12. (3 pts) A car drives 20 miles in 0.4 hours and then drives at a different speed for 0.5 hours. The average speed for the entire trip is 55 mph. Determine the distance the car traveled in the second leg of the trip.

$$29.5 \text{ miles}$$

$$V_{avg} = \frac{d}{t}$$

$$55 \frac{\text{mi}}{\text{hr}} = \frac{20 \text{ mi} + x}{0.4 \text{ hr} + 0.5 \text{ hr}}$$

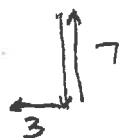
$$x = 29.5 \text{ mi}$$

13. (3 pts) A boat is traveling at 7 ft/s north with respect to a river. The river is flowing at components of 3 ft/s to the west and 7 ft/s to the south. Determine the magnitude of the velocity of the boat with respect to the ground.

$$3 \text{ ft/s}$$

$$\vec{V}_{B/G} = \vec{V}_{B/R} + \vec{V}_{R/G}$$

$$= 7\hat{j} + (-3\hat{i} - 7\hat{j}) = (-3\hat{i}) \text{ ft/s}$$



18. (3 pts) A 90 kg ice hockey player hits a 0.15 kg puck, giving the puck a velocity of 45.0 m/s. If both are initially at rest and if the ice is frictionless, determine the recoil speed of the hockey player.

$$0.075 \text{ m/s}$$

$$m_{HP} v_{HP} + m_P v_P = m_{HP} v'_{HP} + m_P v'_P$$

$$0 + 0 = 90 \text{ kg } v'_{HP} + 0.15 \text{ kg } (45 \text{ m/s})$$

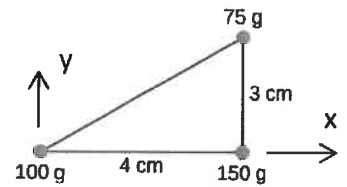
$$v'_{HP} = -0.075 \text{ m/s}$$

19. (3 pts) Three point masses are at the locations shown. Determine the x-coordinate of the center of mass.

$$2.77 \text{ cm}$$

$$\bar{x} = \frac{100 \text{ kg } (0) + 75 \text{ kg } (4 \text{ cm}) + 150 \text{ kg } (4 \text{ cm})}{100 \text{ kg} + 75 \text{ kg} + 150 \text{ kg}} = \frac{900 \text{ kg} \cdot \text{cm}}{325 \text{ kg}}$$

$$= 2.769 \text{ cm}$$

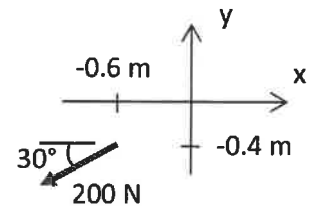


20. (3 pts) A 200 N force acts at (-0.6, -0.4) m. Determine the torque about the origin of the 200 N force. Counter-clockwise is positive.

$$-9.28 \text{ N} \cdot \text{m}$$

$$-200 \text{ N } (\cos 30^\circ) 0.4 \text{ m} + 200 \text{ N } (\sin 30^\circ) (0.6 \text{ m})$$

$$-69.28 + 60 = -9.28 \text{ N} \cdot \text{m}$$



21. (3 pts) A car is going around a 200 ft radius curve. The total acceleration is 18 ft/s^2 , and the car is increasing speed at the rate of 7 ft/s^2 . Determine the speed of the car.

$$57.6 \text{ ft/s}$$

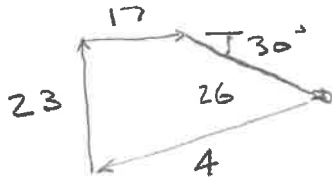
$$a_{\text{tot}} = \sqrt{a_{\text{tan}}^2 + a_n^2} \quad a_n = \frac{v^2}{r}$$

$$18 \frac{\text{ft}}{\text{s}^2} = \sqrt{\left(7 \frac{\text{ft}}{\text{s}^2}\right)^2 + \left(\frac{v^2}{200 \text{ ft}}\right)^2}$$

$$v = 57.59 \text{ ft/s}$$

22. (7 pts) A robot travels 23 m north, 17 m east, and 26 m at 30° south of east. Determine the direction the robot needs to go to return to the starting point.

$$14.2^\circ \text{ S of W}$$



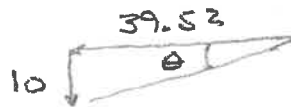
Vector	Magn	Dir	x	y
1	23	$\uparrow$	0	23
2	17	$\rightarrow$	17	0
3	26	$\swarrow 30^\circ$	$26 \cos 30^\circ = 22.52$	$-26 \sin 30^\circ = -13$
4			$\frac{x}{0}$	$\frac{y}{0}$

$$x: 17 + 22.52 + x = 0$$

$$x = -39.52$$

$$y: 23 - 13 - y = 0$$

$$y = -10$$



$$\theta = \tan^{-1}\left(\frac{10}{39.52}\right) = 14.20^\circ$$

23. (7 pts) An amusement park ride starts from rest and drops in free fall for 3.0 seconds. The maximum acceleration that can be used to bring the ride to rest is 65 ft/s^2 . Determine the distance required to bring the ride to rest.

$$71.8 \text{ ft}$$

$$\bullet \downarrow v_1 = 0$$

$$t = 3.0 \text{ s}$$

$$a = -32.2 \text{ ft/s}^2$$

$$\bullet \downarrow v_2$$

$$a = 65 \text{ ft/s}^2$$

$\uparrow$

$$\bullet \downarrow v_3 = 0$$

$$v_2 = v_1 + at$$

$$= 0 + (-32.2 \text{ ft/s}^2)(3.0 \text{ s})$$

$$= -96.6 \text{ ft/s}$$

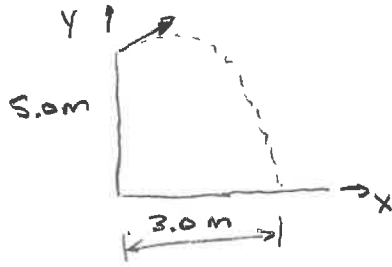
$$s_3 - s_2 = \frac{v_3^2 - v_2^2}{2a} = \frac{0^2 - (-96.6 \text{ ft/s})^2}{2(65 \text{ ft/s}^2)}$$

$$= -71.78 \text{ ft}$$



24. (7 pts) A high diver leaves the end of a 5.0 m high diving board and strikes the water 1.3 seconds later, 3.0 m beyond the end of the board. Determine the magnitude of the initial velocity of the diver.

$$3.42 \text{ m/s}$$



$$V_x = \frac{3.0 \text{ m}}{1.3 \text{ s}} = 2.308 \text{ m/s}$$

$$y_2 = y_1 + V_{1y}t + \frac{1}{2}a_y t^2$$

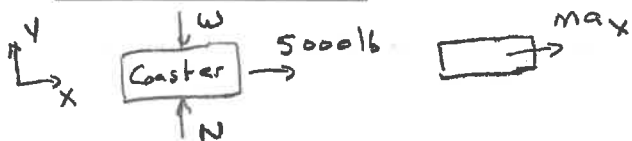
$$0 = 5.0 \text{ m} + V_{1y}(1.3 \text{ s}) + \frac{1}{2}(-9.81 \frac{\text{m}}{\text{s}^2})(1.3 \text{ s})^2$$

$$V_{1y} = 2.530 \text{ m/s}$$

$$V_1^2 = \sqrt{V_{1x}^2 + V_{1y}^2} = \sqrt{(2.308 \text{ m/s})^2 + (2.530 \text{ m/s})^2} = 3.424 \text{ m/s}$$

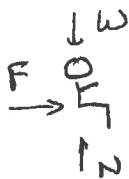
25. (7 pts) The total weight (cars plus riders) of a roller coaster is 9000 lb. A 5000 lb horizontal force is applied to launch the coaster. Determine the horizontal force on a 185 lb rider during the launch. (2 separate FBD = KD required for full credit)

$$103 \text{ lb}$$



$$x: 5000 \text{ lb} = \frac{9000 \text{ lb}}{32.2 \text{ ft/s}^2} a_x$$

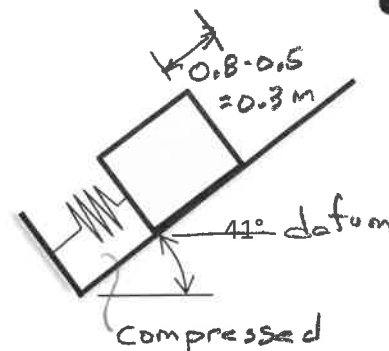
$$a_x = 17.89 \text{ ft/s}^2$$



$$x: F = \frac{185 \text{ lb}}{32.2 \text{ ft/s}^2} (17.89 \text{ ft/s}^2)$$

$$= 102.8 \text{ lb}$$

26. (7 pts) A spring ($k = 120 \text{ N/m}$) has an uncompressed length of 0.8 m . The spring is compressed to a length of 0.5 m and a 2.0 kg block is placed against the spring. The spring is released. Determine the friction force between the block and the inclined surface if the block, attached to the spring, stops just as it reaches the uncompressed position.



$$5.13 \text{ N}$$

$$\frac{1}{2} k x_1^2 + \frac{1}{2} m v_1^2 + m g h_1 + \cancel{U_{s,1}} = \frac{1}{2} k x_2^2 + \frac{1}{2} m v_2^2 + m g h_2 + E_{\text{loss}}$$

$$\frac{1}{2} (120 \frac{\text{N}}{\text{m}}) (0.3 \text{ m})^2 = 2.0 \text{ kg} (9.81 \frac{\text{m}}{\text{s}^2}) 0.3 \text{ m} (\sin 41^\circ) + F (0.3 \text{ m})$$

$$F = 5.128 \text{ N}$$

27. (7 pts) Cart A (3 kg) collides with Cart B (3 kg) which is initially at rest. After the collision, cart B has a velocity of 3.4 m/s . Determine the initial velocity of Cart A if the coefficient of restitution between Cart A and Cart B is 0.8 .



$$3.78 \text{ m/s}$$

$$m_A v_A + m_B v_B = m_A v_A' + m_B v_B'$$

$$\boxed{1} \quad 3 \text{ kg} (v_A) + 3 \text{ kg} (0) = 3 \text{ kg} (v_A') + 3 \text{ kg} (3.4 \text{ m/s})$$

$$e = \frac{-(v_B' - v_A')}{v_B - v_A}$$

$$\boxed{2} \quad 0.8 = \frac{-(3.4 \text{ m/s} - v_A')}{0 - v_A}$$

$$\boxed{1} \quad v_A - v_A' = 3.4$$

$$-0.8 v_A - v_A' = -3.4$$

$$v_A = 3.78 \text{ m/s}$$

$$v_A' = 0.38 \text{ m/s}$$

28. (7 pts) A helicopter rotor blade can be considered to be a long thin rod. Each of the three rotors of a helicopter blade is 3.75 m long and has a mass of 135 kg. Determine the torque that needs to be applied to bring the blades from rest to an angular speed of ~~30~~³⁴ rad/s in 8.0 seconds.

$$8070 \text{ N}\cdot\text{m}$$

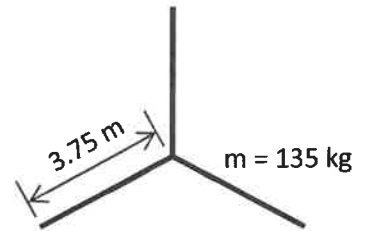
$$I = 3 \left[\frac{1}{12} m l^2 + m d^2 \right]$$

$$= 3 \left[\frac{1}{12} (135 \text{ kg}) (3.75 \text{ m})^2 + 135 \text{ kg} \left(\frac{3.75 \text{ m}}{2} \right)^2 \right]$$

$$= 3 [632.8 \text{ kg}\cdot\text{m}^2] = 1898 \text{ kg}\cdot\text{m}^2$$

$$\alpha = \frac{\omega}{t} = \frac{34 \text{ rad/s}}{8.0 \text{ s}} = 4.25 \text{ rad/s}^2$$

$$\tau = I \alpha = (1898 \text{ kg}\cdot\text{m}^2) (4.25 \frac{\text{rad}}{\text{s}^2}) = 8066 \text{ N}\cdot\text{m}$$



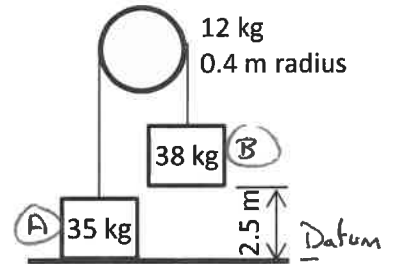
29. (7 pts) Two masses are connected by a rope that hangs over a pulley. The system is initially at rest. Determine the speed of the 38 kg block just before it hits the ground.

$$1.36 \text{ m/s}$$

$$I_{\text{pulley}} = \frac{1}{2} M r^2$$

$$= \frac{1}{2} (12 \text{ kg}) (0.4 \text{ m})^2$$

$$= 0.96 \text{ kg}\cdot\text{m}^2$$



$$\frac{1}{2} m_A v_{A1}^2 + \frac{1}{2} m_B v_{B1}^2 + \frac{1}{2} I \omega_1^2 + m_A g h_{A1} + m_B g h_{B1} = \frac{1}{2} m_A v_{A2}^2 + \frac{1}{2} m_B v_{B2}^2 + \frac{1}{2} I \omega_2^2 + m_A g h_{A2} + m_B g h_{B2}$$

$$38 \text{ kg} (9.81 \frac{\text{m}}{\text{s}^2}) (2.5 \text{ m}) = \frac{1}{2} (35 \text{ kg}) v_2^2 + \frac{1}{2} (38 \text{ kg}) v_2^2 + \frac{1}{2} (0.96 \text{ kg}\cdot\text{m}^2) \left(\frac{v}{0.4 \text{ m}} \right)^2$$

$$+ 35 \text{ kg} (9.81 \frac{\text{m}}{\text{s}^2}) (2.5 \text{ m})$$

$$v_2 = 1.36 \text{ m/s}$$